Attila Szabo

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Hands-on stem cell scientist and lab leader with 8+ years building reliable cell supply for research and cell therapy programs.

Relevant Experience

Calico Life Sciences
Scientific Manager, Cell Culture Mar. 2025 – Present
Principal Research Associate, Cell Culture Core Oct. 2024 – Mar. 2025

- Lead Calico's company-wide **Cell Culture Core** (cell supply, banking and quality testing) for 110+ scientists, nearly doubling its service volume since 2024.
- Built standing production lines for iPSC-derived neurons and stem-cell-derived microglia; the neuron line has supplied 500+ assay-ready plates to 20+ scientists in six labs.
- Improved the neuron process with designed experiments, automated liquid handling and an image-based release assay, nearly doubling neurons per well and cutting seeding variability from 32% to 4% CV.
- Strengthened company-wide cell quality testing: replaced a contamination test that was missing infections, and added karyotyping and pluripotency testing after a seven-vendor evaluation.
- Hire, train and supervise the Core's contract staff, who now run most of its service requests.

Senior Research Associate, Cellular Phenotyping Aug. 2022 – Oct. 2024

- Conceived, built and launched the **Cell Culture Core** company-wide in 2023, designing its request, inventory and reporting systems in Benchling LIMS.
- Automated a cell-based assay to 384-well high-throughput format and ran a small-molecule screen through hit selection.

BlueRock Therapeutics
Research Associate II, Process Development Nov. 2021 – Aug. 2022
Research Associate I, Process Development Mar. 2021 – Nov. 2021
Operations Coordinator, Laboratory Operations Jan. 2018 – Mar. 2021

- Designed and executed technology transfers of stem cell differentiation protocols and multicolor flow cytometry assays.
- Made stem cell differentiation processes more robust using DoE and statistical modelling, and qualified the cell batches that supplied an IND filing.
- Led lab procurement for two sites and 80+ staff, including an ERP roll-out.

Dr. Igor Stagljar Lab, University of Toronto
Undergraduate Research Student, Molecular Genetics May 2016 – Aug. 2017

- Thesis: mapped how the adaptor protein CRKII binds EGFR, a lung cancer drug target, identifying the residues key to binding.

Skills

Cell culture
iPSC and hESC expansion · directed differentiation to dopaminergic neurons, cardiomyocytes and microglia · HSC expansion · primary, cancer and immortalized lines · adherent and suspension culture · bioreactors, cell factories and aseptic pumps

Automation & HTS
Hamilton Vantage (VENUS-trained) · Tecan Fluent · Echo · Certus Flex · BioTek EL406 · Opera Phenix · Incucyte · small-molecule screening

Analytics & quality
High-content imaging · flow cytometry panel design · STR and mycoplasma testing · bulk RNA-seq · DoE · method validation · root-cause investigation

Data & software
Python · SQL · R · Benchling LIMS · Prism · Fiji · FlowJo · Claude Code · Gemini

Selected Projects

SubNetwork
An explorable map of what drugs for neurological and psychiatric conditions act on, built from ChEMBL and the Guide to Pharmacology (subnetwork.dev).

Education

University of Toronto
Honours Bachelor of Science, Molecular Genetics and Biochemistry
2013 – 2017

Certifications

Hamilton VENUS Software Training I
Hamilton Company · Jul. 2025

Machine Learning Specialization
DeepLearning.AI and Stanford Online · May 2025

Lean Six Sigma Black Belt
Six Sigma Global Institute · Sep. 2021

Recognition

Generosity of Spirit and CaliDough Awards
eight peer-nominated Calico awards from six colleagues · 2024 – 2026